SAARLAND UNIVERSITY

Faculty of Humanities
Department of Language Science and Technology
Master Thesis Proposal

Mapping Internal Shifts: A Unified Representational Analysis of Adaptation Methods

submitted by
Khondoker Ittehadul Islam (Matri No: 7085810)
Saarbrücken

Advisor:
Dr. Simon Ostermann
German Research Center for Artificial Intelligence
Saarland Informatics Campus
Saarbrücken, Germany

Reviewer: Arianna Bisazza, PhD

Saarland University
Faculty of Humanities
Department of Language Science and Technology
Campus - Building C7.1
66123 Saarbrücken
Germany

Abstract

Large Language Models are frequently aligned for new tasks using various techniques, ranging from simple prompt adjustments to full model fine-tuning. Currently, researchers evaluate these methods by looking only at the final text the model generates. While this approach tells us what a model can do, it fails to explain how the model's internal mechanics actually change. This research proposal addresses that gap by introducing a unified framework that examines the model's internal hidden states. By viewing all adaptation methods as geometric shifts within the model's internal processing space, this framework enables a direct, consistent comparison of how different techniques alter the model from the inside.

The proposed study evaluates five major adaptation methods: prompting, in-context learning, activation steering, low-rank adaptation, and full fine-tuning. Instead of using separate rules for each technique, the methodology applies the exact same measurements across all of them to track internal movement, structural similarity, and dimensional compression. The project investigates which layers drive the adaptation process, whether different methods eventually reach the same internal subspaces. By uncovering these hidden dynamics, these findings will help researchers detect structural issues and ultimately guide the creation of safer, more efficient, and more transparent language models.

Contents

Chapter 1
Introduction

1.1 Motivation

Large Language Models (LLMs) are commonly adapted to new tasks, domains, and user requirements through a growing collection of techniques. These range from prompting and in-context learning to activation steering, parameter-efficient fine-tuning, and full model fine-tuning. As the number of adaptation strategies continues to increase, a central question emerges: how different are these methods from one another, and do they ultimately alter a model in fundamentally different ways?

Much of the existing literature answers this question by comparing adaptation techniques through downstream performance. Researchers typically evaluate task accuracy, generation quality, computational efficiency, or parameter requirements. While these comparisons are valuable, they provide only an external view of adaptation. They reveal what a model can do after adaptation, but they provide limited insight into how the model internally changes to achieve that behavior.

Recent work has increasingly shifted attention toward internal representations. Rather than evaluating adaptation solely through outputs, researchers examine how adaptation modifies the hidden states of a transformer and how these modifications relate to learning, reasoning, and generalization. This perspective has produced a wide range of findings. Some studies investigate where task-relevant information emerges within a model. Others examine whether adaptation compresses or preserves existing representational structures. Additional work studies how representations evolve during learning, how adaptation influences multilingual behavior, and how internal changes relate to generation quality.

However, the methods used in this literature are often tied to specific adaptation paradigms. For example, certain analyses require direct access to parameter updates and can therefore only be applied to weight-modifying approaches such as fine-tuning. As a result, many existing comparisons cannot be extended uniformly across the full spectrum of adaptation techniques.

1.2 Research Objective

This research is motivated by the unifying framework proposed by Dherin et al. (2025), which argues that the effects of diverse adaptation mechanisms can be expressed through low-rank transformations of a model's internal computations. Under this view, adaptation methods that appear fundamentally different at the implementation level may still produce comparable changes within a shared representational space. This perspective creates an opportunity to compare adaptation techniques using a common analytical framework rather than method-specific evaluation procedures.

Building on this foundation, this study investigates adaptation through the internal representations produced by transformer models. Rather than attempting to reproduce every analytical approach proposed in the literature, this work selects a subset of methods that satisfy two requirements. First, they have demonstrated utility in previous adaptation research. Second, they can be applied consistently across all adaptation mechanisms considered in this study, including prompting, in-context learning, activation steering, low-rank adaptation, and full fine-tuning. This allows the comparison to remain methodologically consistent across adaptation paradigms that operate through different implementation strategies.

The objective of this research is to answer the following questions:

- **RQ1:** How do different adaptation mechanisms alter internal transformer representations?
- **RQ2:** Which transformer layers contribute most strongly to adaptation effects?
- **RQ3:** Do different adaptation mechanisms converge toward similar representational subspaces despite differing implementation strategies?

The findings of this study aim to contribute to two closely related research directions. First, they extend the growing body of representation-based adaptation research that emerged following the framework proposed by Dherin et al. (2025). Second, they provide an empirical perspective on the adaptation taxonomy presented by Ostermann et al. (2026) by investigating whether adaptation categories that differ operationally also differ in the representational changes they induce.

1.3 Paper Roadmap

The remainder of this proposal is organized as follows. Chapter 2 introduces the foundational concepts required throughout the study. The purpose of this chapter is to provide the necessary theoretical background before these concepts are encountered in later chapters. Chapter 3 reviews recent literature that applies these ideas to study adaptation in large language models and summarizes the major empirical findings reported by previous work. Chapter 4 then presents the proposed experimental design. It introduces the adaptation methods and datasets used in this study and describes how each analytical method is employed to address the research questions. Throughout the methodology, the selected analyses are explicitly connected to both the corresponding research questions and the literature findings that motivate their inclusion. Together, these chapters establish a unified framework for investigating how different adaptation mechanisms modify the internal representations of transformer models.

Chapter 2
Background

In this chapter, we introduce the foundational concepts used throughout this thesis. The purpose of this chapter is to provide the structural, geometric, and mathematical background required to understand the representation-based analyses discussed in Chapter 3 and implemented in Chapter 4. The chapter first introduces transformer representations and the residual stream before covering geometric similarity measures, linear probing, dimensionality and spectral metrics, trajectory geometry, and generation evaluation metrics. Together, these concepts provide the analytical vocabulary required to investigate the research questions explored in this study.

2.1 Interpreting Transformer Representations

A central objective of modern interpretability research is understanding how transformers internally encode information. Much of this focuses on the **residual stream**, the primary communication channel between sequential transformer layers. Information generated by attention mechanisms and feed-forward networks is repeatedly written into and read from the residual stream, allowing representations to evolve hierarchically (Elhage et al., 2021).

Early approaches like LogitLens (nostalgebraist, 2020) proved that by projecting intermediate hidden states through the model's pre-trained unembedding matrix, critical semantic information could be decoded early in the network. Subsequent studies confirmed that early layers encode lexical features, middle layers capture syntax, and deeper layers represent abstract semantic reasoning (Dou et al., 2018; Jawahar et al., 2019; Vig and Belinkov, 2019; Feng et al., 2021; Charpentier and Samuel, 2023; Langedijk et al., 2024).

2.2 Linear Separability and Probing

To understand what a model *knows* at a specific layer, researchers utilize **probing** (or diagnostic classification). Probing tests whether a specific concept or feature is linearly accessible within the model's internal representations.

If a concept is **linearly separable**, it means a simple linear hyperplane can divide the representations into distinct target classes. To measure this, a linear probing layer is trained on frozen hidden states h_l at a specific layer l. The classifier computes a predicted probability y using a learned weight matrix W and bias b:

$$y = \sigma(Wh_l + b)$$

Where σ is a standard activation function (e.g., softmax or sigmoid). High probing accuracy indicates that the model has explicitly encoded the target feature in a readily extractable, linear format at that specific depth. Because training a simple logistic regression probe requires minimal compute, mapping the linear separability of concepts across all layers of an LLM is much more efficient for experiments.

2.3 Dimensionality and Spectral Metrics

Adaptation methods do not just move representations; they alter the actual shape and volume of the latent space. To quantify this structural compression or expansion, existing literature (Section 3.2, 3.3) analyzed the variance of hidden states using **Eigenvalue Decomposition**.

For a matrix of hidden states H, we compute the covariance matrix C. Decomposing C yields a set of eigenvalues λ_i, representing the variance captured along each principal component. From these eigenvalues, two vital metrics of complexity are derived:

- **Intrinsic Dimensionality (ID):** Measures the minimum number of dimensions required to capture the essential geometric structure of the data. A common estimator is the Participation Ratio, defined as:

$$\text{ID} = \frac{(\sum \lambda_i)^2}{\sum \lambda_i^2}$$

- **Effective Rank (ER):** Evaluates the uniformity of the variance distribution across dimensions using Shannon entropy. It indicates how many dimensions are meaningfully contributing to the representation:

$$\text{ER} = \exp\left(-\sum p_i \ln p_i\right)$$

  where $p_i = \frac{\lambda_i}{\sum \lambda_j}$ is the normalized variance of the i-th dimension

Calculating these metrics involves straightforward Principal Component Analysis (PCA) and singular value decomposition algorithms. These metrics will allow mapping out representation spaces.

2.4 Trajectory Geometry

Viewing reasoning and generation as continuous movements through representation space, we can map the step-by-step evolution of a hidden state: $h_1 \to h_2 \to h_3 \to \ldots$.

Existing Literature (Section 3.4) evaluated this dynamic path using **Trajectory Curvature (**κ**)**, which measures the deviation of the representation path from a straight line. Low curvature implies direct, stable progression, while high curvature indicates uncertainty or representation instability.

Chapter 3
Literature Review

With the foundational concepts established in Chapter 2, this chapter reviews how these methods have been applied in recent literature to study adaptation in large language models. The purpose of this review is to show how researchers use representation-based analyses to investigate changes in internal model behavior beyond downstream performance metrics. The chapter covers studies on knowledge localization, representational collapse, spectral geometry, representational trajectories, and cross-lingual representations. These works provide the empirical foundation for the research questions explored in this thesis and motivate the analytical framework presented in Chapter 4.

3.1 Uncovering Knowledge via Linear Probing

The adoption of **linear separability** and probing layers revolutionized how researchers track concept emergence. Originally popularized by Alain and Bengio (2016), diagnostic classifiers have become the standard for locating where a model solves a task internally.

Recent literature heavily utilizes these probes to compare adaptation methods. For instance, the TaskTracker framework (Abdelnabi et al., 2025) examined activation differences when a model transitions between competing objectives.

By calculating activation deltas ($\Delta h_l = h_l^{\text{adapted}} - h_l^{\text{base}}$), researchers trained linear probes and found that these displacements become linearly separable within early-to-mid layers. This proved that different adaptation methods generate robust, linearly decipherable signals well before the final output layer.

3.2 Representational Collapse and Dimensional Compression

A central question in recent literature is whether adaptation compresses the original, rich pre-training geometry of a model. To answer this, researchers turned to the spectral metrics of **Intrinsic Dimensionality** and **Effective Rank** (From Section 2.3).

Historically, fine-tuned models performed well on targeted tasks but transferred poorly to adjacent domains. Investigations by Doimo et al. (2024) and Janapati and Ji (2025) applied Intrinsic Dimensionality (Section 2.3) calculations to pre- and post-adapted models. They empirically confirmed the phenomenon of **representational collapse**: under full fine-tuning, the Effective Rank (Section 2.3) of the latent space sharply decreases. The model discards irrelevant pre-training geometry, *condensing its variance into a much narrower functional subspace*. This literature validates that measuring eigenvalues is a highly effective strategy to predict a model's loss of generalizability.

3.3 Spectral Geometry: LoRA vs. Full Fine-Tuning

Parameter-efficient fine-tuning (PEFT), specifically LoRA, is frequently evaluated against full fine-tuning (FFT). Shuttleworth et al. (2026) utilized **Eigenvalue Decomposition** (Section 2.3) to analyze the spectral geometry of both methods, proving that identical downstream accuracy does not imply identical internal structures.

By examining the principal components of the adapted latent spaces, they discovered that LoRA frequently introduces "intruder dimensions"—directions of high variance that successfully support task performance but fundamentally *alter the original geometric organization*. FFT, conversely, tends to operate within the existing dimensional structures.

3.4 Representational Trajectories and In-Context Learning

Moving from static states to temporal progression, recent work has examined how representations evolve dynamically. Xiong et al. (2026) demonstrated that successful in-context learning (ICL) is associated with the progressively *increasing linear separability of task-relevant concepts as the context window scales*.

Building on this, Sun et al. (2026) applied **Trajectory Curvature** (Section 2.4) metrics to model reasoning. They found that correct reasoning steps exhibit low curvature (direct paths), while incorrect reasoning follows high-curvature paths full of redirection. This literature shifts the paradigm, proving that *simple geometric path analysis can reliably predict hallucination* and uncertainty without needing labeled ground-truth data (Chen et al., 2025).

3.5 Structural Stability

Finally, evaluations using open-ended generation tasks demonstrate the delicate balance of adaptation. Braun et al. (2026) utilized the `Distinct-2 word` metric alongside internal structural measurements to assess activation steering. They found a strict trade-off: forcing strong geometric interventions (high steering coefficients) improves targeted concept control, but results in a sharp collapse in `Distinct-2 word` scores, *manifesting as token loops and degraded fluency*.

Chapter 4
Methodology

Building on the concepts introduced in Chapter 2 and the findings reviewed in Chapter 3, this chapter presents the methodology used to investigate adaptation-induced representational changes in large language models. The purpose of this chapter is to describe how each research question is operationalized through a unified activation-space framework. The chapter first introduces the research questions, adaptation methods, and evaluation environments. It then details the analyses used to measure hidden-state displacement, representational similarity, dimensionality, spectral geometry, layer importance, representational trajectories, and cross-lingual alignment. Together, these procedures provide a systematic framework for comparing how different adaptation mechanisms modify internal transformer representations.

4.1 Research Design

The objective of this study is to compare how different adaptation mechanisms alter the internal representations of large language models. Building on the representation-based interpretability framework established in the literature review, our analysis focuses on activation-space changes rather than solely on downstream task performance.

This methodology adopts a unified activation-space framework. Because distinct adaptation techniques can be expressed mathematically as low-rank transformations within the hidden-state space, we can map and compare them through the explicit geometric shifts they induce (Dherin et al., 2025).

4.2 Research Questions

Our empirical design is organized around four primary research questions:

- **RQ1:** How do different adaptation mechanisms alter internal transformer representations?

- **RQ2:** Which transformer layers contribute most strongly to adaptation effects?
- **RQ3:** Do different adaptation mechanisms converge toward similar representational subspaces despite differing implementation strategies?

4.3 Adaptation Methods

To capture the full adaptation spectrum and trace specific structural shifts, we evaluate five mechanisms:

- **External Steering Adaptation**: Prompting and In-Context Learning (ICL).
- **Activation-Space Steering Adaptation:** Difference-Mean Activation Steering (Marks and Tegmark, 2023).
- **Parameter-Space Adaptation:** Low-Rank Adaptation (LoRA) (Hu et al., 2022).

4.4 Experimental Setup

4.4.1 Task Environments

To select an appropriate benchmark, we considered four criteria. First, the task should require multi-word responses rather than single-token outputs. Second, the answer should be generated from the information provided in the question without requiring explicit external knowledge. Third, the task should not depend on an additional passage or supporting context. Finally, our selected Adaptation Methods (Sec 4.3) could be adapted in the benchmark, i.e., sufficient training instances for LoRA adapters. etc.,

Based on these criteria, we select the following benchmark:

- **Evil Trait task**: Chen et al. (2025) derive Difference-Mean steering vectors for three personality traits using 20 question-response pairs, each consisting of a common trait-induced system prompt (positive) and a corresponding contrast (negative). In addition, they release approximately 5,000 question-response pairs for each variant (Normal, Mistake I, and Mistake II) and use them to train LoRA adapters for comparison with steering.

For our experiments, we focus on the *Evil* trait and reuse the released steering vector, the corresponding positive system prompt for prompt-based adaptation, and 5 *Evil* trait responses generated by the model as demonstrations for ICL. For LoRA, we fine-tune on the `Mistake II` dataset, as this variant demonstrated performance comparable to Difference-Mean steering in the original paper. All adaptation methods are then evaluated on the authors' 20 evaluation questions using the corresponding evaluation prompt. To avoid API credits, we use three open-source LLM-as-a-Judge models and report the average trait score.

4.4.2 Models

To evaluate whether adaptation-induced representational changes are consistent across model families, we select two open-source language models of comparable size that have been used in our selected benchmark. All models are decoder-only transformers trained using an autoregressive next-token prediction objective, but they differ in their pre-training data and input representation strategies.

- **Llama-3.1-8B-Instruct** (Grattafiori et al., 2024): a decoder-only transformer consisting of 32 layers and trained on a large collection of web, code, and instructional data.
- **Qwen2.5-7B-Instruct** (Hui et al., 2024): a decoder-only transformer consisting of 28 layers and trained on a large multilingual corpus.

For the respective LLM-as-a-Judge models, we choose the following large parameterized open-source models to replicate the similar performance of closed models:

- Llama-3.3-70B-Instruct
- Qwen-2.5-72B-Instruct
- Llama-3.1-70B-Instruct

4.5 Unified Activation-Space Framework

For an input sequence x, we extract hidden representations from the residual stream of every transformer layer. Let h_l^{base} denote the hidden representation produced by the base model at layer l, and h_l^{adapted} denote the corresponding representation after adaptation.We quantify adaptation uniformly across all methods through layer-wise activation displacement:

$$\Delta h_l = h_l^{\text{adapted}} - h_l^{\text{base}}$$

To ensure comparability across prompting and ICL configurations with differing context lengths, we compute all measurements using aligned target-token representations rather than absolute sequence positions.

4.6 Hidden-State Displacement Analysis

Addresses RQ1, RQ2; relates to Section 3.1, i.e., Uncovering Knowledge via Linear Probing.

The first stage of our analysis examines how strongly each adaptation method shifts internal representations. For every layer l, we compute the magnitude of representational displacement:

$$||\Delta h_l||_2$$

Additionally, we calculate the directional alignment between adaptation vectors from different methods (A and B) using cosine similarity:

$$\text{Cosine}(\Delta h_l^A, \Delta h_l^B)$$

Using the linear probing techniques outlined in Section 2.2, we will assess these displacements to identify precisely where task drift becomes linearly separable within the network.

4.7 Representational Collapse and Dimensionality Analysis

Addresses RQ1, RQ3; relates to Section 3.2, i.e., Representational Collapse and Dimensional Compression.

While the literature predominantly applies Intrinsic Dimensionality and Effective Rank to measure representational collapse in FFT, our methodology extends these metrics across all five adaptation techniques.

To onboard methods that do not alter weights (like ICL and Prompting), we compute the covariance matrix over a standardized batch of input representations at layer l for each adapted state. We then extract the eigenvalues to calculate the Effective Rank (ER) and Intrinsic Dimensionality (ID). By directly comparing these dimensionality metrics across paradigms—for instance, evaluating $\mathrm{ER}(h_l^{\mathrm{FFT}})$ versus $\mathrm{ER}(h_l^{\mathrm{ICL}})$—we quantitatively verify whether context-based adaptation compresses pre-training geometry in the same manner as gradient-based fine-tuning.

4.8 Spectral Geometry Analysis

Addresses RQ1, RQ3; relates to Section 3.3, i.e., Spectral Geometry: LoRA vs. Full Fine-Tuning.

To address the potential emergence of "intruder dimensions" identified in parameter-efficient adaptation, we conduct a spectral analysis using Eigenvalue Decomposition (Section 2.3).

Similar to our dimensionality analysis, we apply this uniformly across all methods. By comparing the principal component energy distribution of Δh_l generated by LoRA against those generated by Steering, ICL, and Prompting, we can explicitly identify whether high-variance anomaly dimensions are exclusive to parameter updates or if they also manifest during purely contextual or activation-based interventions.

4.9 Layer Importance and Causal Analysis

Addresses RQ2; relates to Section 3.1, i.e., Uncovering Knowledge via Linear Probing.

Consistent with findings that transformer representations are organized hierarchically, we identify where adaptation fundamentally occurs. For each layer, we correlate the displacement magnitude with task performance improvements to estimate adaptation-sensitive depths

$$\mathrm{Importance}(l) = \mathrm{Corr}(||\Delta h_l||_2, \mathrm{Performance})$$

4.10 Representational Trajectory Analysis

Addresses RQ1 & RQ2; relates to Section 3.4, & 3.5, i.e., Representational Trajectories and In-Context Learning & Structural Stability.

We model adaptation as a sequence of representational states by progressively increasing intervention strength. To ensure all adaptation techniques can be analyzed via Trajectory Length and Trajectory Curvature, we define the temporal step variable t according to the specific mechanism:

- **ICL**: t represents the number of demonstration shots (0-shot to k-shots).
- **Prompting**: t represents sequential levels of prompt refinement or specificity.
- **Steering**: t represents the incrementally increasing steering coefficient.
- **LoRA/FFT**: t represents the training checkpointsR across epochs.

By tracking h_t across these standardized steps, we compute curvature and length to reveal whether an adaptation method moves directly toward a stable subspace or exhibits instability. Furthermore, we train linear probes across t to identify the emergence depth of statistically significant task-relevant concepts.

- **Cosine Alignment:** Measuring directional similarity via Cosine$(\Delta h_l^{(i)}, \Delta h_l^{(j)})$

4.11 Experimental Pipeline

The complete experimental procedure consists of seven sequential and modular stages:

1. Run the base model and extract baseline hidden representations on the evaluation set of *Evil Trait*.
2. Apply each adaptation method and extract the adapted hidden representations (h_l^{adapted}).
3. Compute the layer-wise activation displacements (Δh_l).
4. Evaluate representational similarity, calculate Effective Rank, and extract eigenvalues for spectral geometry analysis across all mapped techniques.
5. Identify causally important layers through performance correlation and linear probing.
6. Construct adaptation trajectories using standard temporal steps (t) to analyze length, and curvature alignment.

Chapter 5
Conclusion

Large Language Models adapt to new tasks through many different methods. Most research evaluates these methods by looking only at the final text the model produces. This proposal takes a different approach by examining the internal hidden states of the models. By treating all adaptation methods as geometric changes within the model's internal space, this study introduces a unified framework to compare them directly. This allows us to understand what happens inside the model during adaptation, rather than just scoring its final output.

The experimental design tests five specific methods: prompting, in-context learning, activation steering, low-rank adaptation, and full fine-tuning. Instead of using different evaluation rules for each method, the methodology applies the exact same mathematical measurements across the board. These core measurements track hidden-state displacement, representational similarity, and dimensional compression. This consistent testing environment allows us to see exactly how different techniques alter the internal structures of the transformer model.

Through this experimental pipeline, the study directly addresses its four main research questions. It identifies which specific layers drive model adaptation and measures how strongly internal representations change. The experiments also test whether fundamentally different adaptation methods eventually converge on the same internal target spaces.

Ultimately, this research moves beyond standard performance benchmarks to reveal how models actually change inside. The findings will provide a clear comparison between methods that alter user context and methods that alter model weights. They will also help identify structural issues, such as representational collapse, across all adaptation types. By providing a solid empirical foundation for how adaptation functions internally, this study aims to guide future efforts in building safer, more transparent, and more efficient language models.

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